

☐ STAAD SPACE

 STRUCTURIX	DUBAI METRO PROJECT		Sheet:
	RTA		
	REV_A3	CAR PARKING DESIGN TYPE P1 TO P3	

41. 35 34 35; 36 35 36; 37 36 4; 38 2 37; 39 37 38; 40 38 39; 41 39 40; 42 40 41
 42. 43 41 53; 44 42 43; 45 43 44; 46 44 45; 47 45 46; 48 46 47; 49 47 48; 50 48 49
 43. 51 49 50; 52 50 51; 53 51 5; 54 52 27; 55 53 42; 56 54 16; 76 6 64; 77 5 66
 44. 78 53 71; 79 52 69; 80 4 68; 81 54 65; 83 64 74; 84 64 65; 85 65 75; 86 66 76
 45. 88 68 78; 89 69 79; 90 68 69; 91 66 71; 92 71 77; 93 65 80; 94 64 81; 95 71 82
 46. 96 66 83; 98 74 60; 99 75 57; 101 74 75; 102 76 59; 103 77 56; 104 76 77
 47. 105 78 58; 106 79 55; 107 78 79; 108 80 71; 109 54 80; 110 80 53; 111 81 66
 48. 112 6 81; 113 81 5; 114 82 69; 115 53 82; 116 82 52; 117 83 68; 118 5 83
 49. 119 83 4
 50. DEFINE MATERIAL START
 51. ISOTROPIC STEEL
 52. E 2.05E+008
 53. POISSON 0.3
 54. DENSITY 76.8195
 55. ALPHA 1.2E-005
 56. DAMP 0.03
 57. END DEFINE MATERIAL
 58. CONSTANTS
 59. MATERIAL STEEL MEMB 1 TO 4 6 TO 56 76 TO 81 83 TO 86 88 TO 96 98 99 101 TO 119
 60. MEMBER PROPERTY AMERICAN
 61. 76 77 80 TAPERED 0.15 0.006 0.1897 0.15 0.01 0.15 0.01
 62. 83 86 88 TAPERED 0.1897 0.006 0.2919 0.15 0.01 0.15 0.01
 63. 78 79 81 TAPERED 0.15 0.006 0.1778 0.15 0.01 0.15 0.01
 64. 85 89 92 TAPERED 0.1778 0.006 0.2904 0.15 0.01 0.15 0.01
 65. MEMBER PROPERTY BRITISH
 66. 6 TO 56 TABLE ST PIPE OD 0.1683 ID 0.15408
 67. 93 TO 96 108 111 114 117 TABLE ST TUBE TH 0.0045 WT 0.15 DT 0.15
 68. 1 TO 4 TABLE ST PIPE OD 0.1143 ID 0.10226
 69. 84 90 91 101 104 107 TABLE ST TUBE TH 0.0045 WT 0.15 DT 0.15
 70. MEMBER PROPERTY BRITISH
 71. 98 102 105 TAPERED 0.2919 0.006 0.3 0.15 0.01 0.15 0.01
 72. MEMBER PROPERTY BRITISH
 73. 99 103 106 TAPERED 0.2904 0.006 0.3 0.15 0.01 0.15 0.01
 74. 109 110 112 113 115 116 118 119 TABLE ST PIPE OD 0.0603 ID 0.05248
 75. SUPPORTS
 76. 55 TO 60 PINNED
 77. MEMBER RELEASE
 78. 1 TO 4 START MX MY MZ
 79. 1 TO 4 END MX MY MZ
 80. 93 TO 96 START MX MY MZ
 81. 108 111 114 117 END MX MY MZ
 82. 109 110 112 113 115 118 START MX MZ
 83. 109 110 112 113 116 119 END MX MY MZ
 84. LOAD 1 PRESTRESS
 85. JOINT LOAD
 86. 1 FX 12.5051 FY 0.754175 FZ 2.49874
 87. 2 FX 0 FY 1.81295 FZ 5.95756
 88. 3 FX -12.5051 FY 0.754175 FZ 2.49874
 89. 4 FX -12.4102 FY -0.243969 FZ -2.49498
 90. 5 FX 0 FY -0.560719 FZ -5.9718
 91. 6 FX 12.4102 FY -0.243969 FZ -2.49498
 92. 7 FX 0.335808 FY -0.020436 FZ -0.010948
 93. 8 FX 0.250016 FY -0.018263 FZ 0.000486
 94. 9 FX 0.418385 FY -0.039849 FZ 0.010388
 95. 10 FX 0.330118 FY -0.040641 FZ -0.014803
 96. 11 FX 0.244377 FY -0.02846 FZ 0.014684

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 structurex P.O. Box 19999 Dubai, UAE Phone: 04 3911111 Fax: 04 3911112 Email: info@structurex.ae Website: www.structurex.ae	DUBAI METRO PROJECT		Sheet:
	RTA		
	REV_A3	CAR PARKING DESIGN TYPE P1 TO P3	

97. 12 FX 0.426718 FY -0.056974 FZ 7.2E-005
 98. 13 FX 0.313643 FY -0.044456 FZ -0.003994
 99. 14 FX 0.234835 FY -0.032058 FZ 0.004305
 100. 15 FX 0.452198 FY -0.067862 FZ -0.015449
 101. 16 FX 0.276059 FY -0.035498 FZ 0.007106
 102. 17 FX 0.224563 FY -0.027599 FZ 0.009293
 103. 18 FX 0.496933 FY -0.06059 FZ 0.000689
 104. 19 FX 0.210503 FY -0.019617 FZ 0.000342
 105. 20 FX 0.295352 FY -0.0281 FZ -0.00149
 106. 21 FX 0.407535 FY -0.018161 FZ 0.002727
 107. 22 FX -0.407535 FY -0.018161 FZ 0.002727
 108. 23 FX -0.295352 FY -0.0281 FZ -0.00149
 109. 24 FX -0.210503 FY -0.019617 FZ 0.000342
 110. 25 FX -0.496933 FY -0.06059 FZ 0.000689
 111. 26 FX -0.224563 FY -0.027599 FZ 0.009293
 112. 27 FX -0.276059 FY -0.035498 FZ 0.007106
 113. 28 FX -0.452198 FY -0.067862 FZ -0.015449
 114. 29 FX -0.234835 FY -0.032058 FZ 0.004305
 115. 30 FX -0.313643 FY -0.044456 FZ -0.003994
 116. 31 FX -0.426718 FY -0.056974 FZ 7.2E-005
 117. 32 FX -0.244377 FY -0.02846 FZ 0.014684
 118. 33 FX -0.330118 FY -0.040641 FZ -0.014803
 119. 34 FX -0.418385 FY -0.039849 FZ 0.010388
 120. 35 FX -0.250016 FY -0.018263 FZ 0.000486
 121. 36 FX -0.335808 FY -0.020436 FZ -0.010948
 122. 37 FX 0 FY -0.050155 FZ -8E-006
 123. 38 FX 0 FY -0.066194 FZ 1.3E-005
 124. 39 FX 0 FY -0.053439 FZ 6E-006
 125. 40 FX 0 FY -0.116905 FZ 3.7E-005
 126. 41 FX 0 FY -0.072502 FZ 5.2E-005
 127. 42 FX 0 FY -0.078196 FZ 3.1E-005
 128. 43 FX 0 FY -0.128859 FZ -4.8E-005
 129. 44 FX 0 FY -0.074394 FZ -5.9E-005
 130. 45 FX 0 FY -0.09005 FZ -3.6E-005
 131. 46 FX 0 FY -0.116829 FZ -4.7E-005
 132. 47 FX 0 FY -0.069968 FZ -3.7E-005
 133. 48 FX 0 FY -0.084302 FZ -4E-006
 134. 49 FX 0 FY -0.088827 FZ -1E-006
 135. 50 FX 0 FY -0.050632 FZ 3.5E-005
 136. 51 FX 0 FY -0.054324 FZ 2.9E-005
 137. SELFWEIGHT Y -1
 138. LOAD 2 WIND DN
 139. JOINT LOAD
 140. 1 FX 14.4413 FY 0.255477 FZ 2.94039
 141. 2 FX 0.007727 FY 1.01283 FZ 7.47016
 142. 3 FX -14.4316 FY 0.247277 FZ 2.94286
 143. 4 FX -12.1807 FY -0.750721 FZ -2.22066
 144. 5 FX -0.002117 FY -1.55942 FZ -4.82018
 145. 6 FX 12.1804 FY -0.753788 FZ -2.21852
 146. 7 FX 1.37334 FY -0.239353 FZ 0.013884
 147. 8 FX 1.22821 FY -0.238884 FZ 0.024031
 148. 9 FX 2.75543 FY -0.542564 FZ 0.062265
 149. 10 FX 2.39865 FY -0.51677 FZ 0.032094
 150. 11 FX 1.78258 FY -0.39373 FZ 0.051095
 151. 12 FX 3.14974 FY -0.685041 FZ 0.060793
 152. 13 FX 2.31235 FY -0.524527 FZ 0.043835
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 structureflex P.O. Box 12345 Dubai, U.A.E. Tel: +971 4 123 4567 Fax: +971 4 123 4568 Email: info@structureflex.ae	DUBAI METRO PROJECT		Sheet:
	RTA		
	REV_A3	CAR PARKING DESIGN TYPE P1 TO P3	

153. 14 FX 1.73365 FY -0.399924 FZ 0.042413
 154. 15 FX 3.36903 FY -0.750574 FZ 0.053101
 155. 16 FX 2.0638 FY -0.463069 FZ 0.056131
 156. 17 FX 1.67834 FY -0.380415 FZ 0.051259
 157. 18 FX 3.63149 FY -0.762927 FZ 0.08714
 158. 19 FX 1.26326 FY -0.260484 FZ 0.04088
 159. 20 FX 1.76694 FY -0.352958 FZ 0.052563
 160. 21 FX 0.845583 FY -0.123724 FZ 0.02733
 161. 22 FX -0.863584 FY -0.134383 FZ 0.030397
 162. 23 FX -1.76674 FY -0.354281 FZ 0.052644
 163. 24 FX -1.26312 FY -0.261412 FZ 0.040933
 164. 25 FX -3.63129 FY -0.763367 FZ 0.086638
 165. 26 FX -1.67833 FY -0.380498 FZ 0.051014
 166. 27 FX -2.06378 FY -0.463172 FZ 0.05583
 167. 28 FX -3.36902 FY -0.750663 FZ 0.052634
 168. 29 FX -1.73366 FY -0.399949 FZ 0.042186
 169. 30 FX -2.31236 FY -0.524554 FZ 0.043537
 170. 31 FX -3.14977 FY -0.685035 FZ 0.060421
 171. 32 FX -1.78263 FY -0.393678 FZ 0.050895
 172. 33 FX -2.39875 FY -0.516671 FZ 0.031832
 173. 34 FX -2.75627 FY -0.54137 FZ 0.061908
 174. 35 FX -1.22716 FY -0.233056 FZ 0.023525
 175. 36 FX -1.37089 FY -0.228113 FZ 0.013009
 176. 37 FX -8E-006 FY -0.289926 FZ 5E-006
 177. 38 FX -9E-006 FY -0.749816 FZ 6E-006
 178. 39 FX -6E-006 FY -0.561592 FZ 0
 179. 40 FX 8E-006 FY -1.4565 FZ 7E-006
 180. 41 FX -7E-006 FY -0.793832 FZ 6E-006
 181. 42 FX -7E-006 FY -0.86675 FZ 5E-006
 182. 43 FX 1E-006 FY -1.46144 FZ 7E-006
 183. 44 FX -1.2E-005 FY -0.769123 FZ 7E-006
 184. 45 FX -1.9E-005 FY -0.967188 FZ 3E-006
 185. 46 FX 2.1E-005 FY -1.32801 FZ 8E-006
 186. 47 FX 1E-005 FY -0.753756 FZ 3E-006
 187. 48 FX 1.2E-005 FY -0.967692 FZ 1.2E-005
 188. 49 FX -1E-005 FY -1.05489 FZ 8E-006
 189. 50 FX -5E-006 FY -0.454829 FZ 3E-006
 190. 51 FX -6E-006 FY -0.45581 FZ 2E-006
 191. SELFWEIGHT Y -1
 192. LOAD 3 WIND UP
 193. JOINT LOAD
 194. 1 FX 41.596 FY 4.7184 FZ 8.45871
 195. 2 FX 0.007949 FY 9.19338 FZ 16.6535
 196. 3 FX -41.5883 FY 4.73721 FZ 8.45848
 197. 4 FX -41.1485 FY 1.1217 FZ -9.20783
 198. 5 FX -0.003503 FY 2.05837 FZ -20.2354
 199. 6 FX 41.1441 FY 1.13203 FZ -9.21678
 200. 7 FX 0.679009 FY 0.099195 FZ -0.0333
 201. 8 FX 0.602423 FY 0.104088 FZ -0.021512
 202. 9 FX 1.32531 FY 0.199304 FZ -0.040938
 203. 10 FX 1.34976 FY 0.235135 FZ -0.07829
 204. 11 FX 1.03545 FY 0.193868 FZ -0.03801
 205. 12 FX 1.94261 FY 0.325868 FZ -0.102504
 206. 13 FX 1.50345 FY 0.272701 FZ -0.094566
 207. 14 FX 1.11996 FY 0.212786 FZ -0.066612
 208. 15 FX 2.09675 FY 0.345225 FZ -0.147003
 □ STAAD SPACE

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 101, DUNE, P.O. BOX 200, DUBAI, U.A.E. TEL: 04 381 1111 FAX: 04 381 1111 E-MAIL: info@structurflex.ae WWW.STRUCTURFLEX.AE	DUBAI METRO PROJECT		Sheet:
	RTA		
	REV_A3	CAR PARKING DESIGN TYPE P1 TO P3	

209. 16 FX 1.20625 FY 0.217464 FZ -0.075317
 210. 17 FX 0.981842 FY 0.18459 FZ -0.060427
 211. 18 FX 1.7767 FY 0.265213 FZ -0.110336
 212. 19 FX 0.560289 FY 0.099621 FZ -0.037309
 213. 20 FX 0.778159 FY 0.123897 FZ -0.050301
 214. 21 FX 0.438002 FY 0.053318 FZ -0.013762
 215. 22 FX -0.453844 FY 0.058608 FZ -0.015904
 216. 23 FX -0.778495 FY 0.125396 FZ -0.051214
 217. 24 FX -0.56052 FY 0.100671 FZ -0.037955
 218. 25 FX -1.77626 FY 0.265551 FZ -0.11129
 219. 26 FX -0.981777 FY 0.184518 FZ -0.060857
 220. 27 FX -1.20617 FY 0.217376 FZ -0.075845
 221. 28 FX -2.09672 FY 0.344995 FZ -0.147883
 222. 29 FX -1.11998 FY 0.212662 FZ -0.067081
 223. 30 FX -1.50348 FY 0.272534 FZ -0.095197
 224. 31 FX -1.94279 FY 0.325577 FZ -0.103337
 225. 32 FX -1.03573 FY 0.193519 FZ -0.03845
 226. 33 FX -1.35023 FY 0.234556 FZ -0.078862
 227. 34 FX -1.32632 FY 0.196532 FZ -0.041381
 228. 35 FX -0.599206 FY 0.09963 FZ -0.021533
 229. 36 FX -0.672141 FY 0.091338 FZ -0.033165
 230. 37 FX 0.000189 FY 0.14687 FZ 2.6E-005
 231. 38 FX -1.9E-005 FY 0.325908 FZ 6.2E-005
 232. 39 FX -1.8E-005 FY 0.2544 FZ -0.000372
 233. 40 FX 0 FY 0.669138 FZ -9.7E-005
 234. 41 FX 4.5E-005 FY 0.465297 FZ 0.000135
 235. 42 FX 4.9E-005 FY 0.500463 FZ -0.000242
 236. 43 FX -1.4E-005 FY 0.873751 FZ -9.2E-005
 237. 44 FX -9E-006 FY 0.484289 FZ 5E-005
 238. 45 FX -1.2E-005 FY 0.607914 FZ -0.000135
 239. 46 FX -1.9E-005 FY 0.762291 FZ 2.8E-005
 240. 47 FX -1.4E-005 FY 0.402705 FZ -4E-005
 241. 48 FX -2E-005 FY 0.497801 FZ -5.1E-005
 242. 49 FX 1E-005 FY 0.415891 FZ 8E-005
 243. 50 FX 5.1E-005 FY 0.178774 FZ 1.2E-005
 244. 51 FX 9.7E-005 FY 0.165907 FZ -8.5E-005
 245. SELFWEIGHT Y -1
 246. LOAD COMB 101 1.4PRESTRESS
 247. 1 1.4
 248. LOAD COMB 102 1.4 WIND DN
 249. 2 1.4
 250. LOAD COMB 103 1.4 WIND UP
 251. 3 1.4
 252. PERFORM ANALYSIS
☐ STAAD SPACE

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 STRUCTURFLEX 10, Avenue des Nations 1200, Geneva Suisse Tél. 022 733 11 11 Telex 240000 Fax 022 733 11 12	DUBAI METRO PROJECT		Sheet:
	RTA		
	REV_A3	CAR PARKING DESIGN TYPE P1 TO P3	

PROBLEM STATISTICS

NUMBER OF JOINTS/MEMBER+ELEMENTS/SUPPORTS = 76/ 95/ 6
 ORIGINAL/FINAL BAND-WIDTH= 72/ 10/ 63 DOF
 TOTAL PRIMARY LOAD CASES = 3, TOTAL DEGREES OF FREEDOM = 438
 SIZE OF STIFFNESS MATRIX = 28 DOUBLE KILO-WORDS
 REQRD/AVAIL. DISK SPACE = 12.4/ 31546.5 MB, EXMEM = 400.2 MB

253. PARAMETER

254. CODE BS5950

255. LY 3.5 MEMB 12 TO 21 28 TO 37 44 TO 56

256. LZ 3.5 MEMB 12 TO 21 28 TO 37 44 TO 56

257. UNL 3.5 MEMB 12 TO 21 28 TO 37 44 TO 56

258. LY 2.2 MEMB 6 TO 11 22 TO 27 38 TO 43

259. LZ 2.2 MEMB 6 TO 11 22 TO 27 38 TO 43

260. UNL 2.2 MEMB 6 TO 11 22 TO 27 38 TO 43

261. CHECK CODE MEMB 1 TO 4 6 TO 56 76 TO 81 83 TO 86 88 TO 96 98 99 101 TO 119

STEEL DESIGN

STAAD.Pro CODE CHECKING - (BSI)

PROGRAM CODE REVISION V2.9_5950-1_2000

□ STAAD SPACE

ALL UNITS ARE - KN METE (UNLESS OTHERWISE NOTED)

MEMBER	TABLE	RESULT/	CRITICAL COND/	RATIO/	LOADING/
	FX	MY	MZ	LOCATION	

=====

1 ST PIPE	PASS	BS-4.7 (C)	0.274	103	
57.65 C		0.00	0.00	0.00	
2 ST PIPE	PASS	BS-4.7 (C)	0.274	103	
57.64 C		0.00	0.00	0.00	
3 ST PIPE	PASS	BS-4.7 (C)	0.257	103	
54.14 C		0.00	0.00	0.00	
4 ST PIPE	PASS	BS-4.7 (C)	0.257	103	
54.18 C		0.00	0.00	0.00	
6 ST PIPE	PASS	ANNEX I.1	0.014	102	
3.74 C		0.12	0.60	-	
7 ST PIPE	PASS	ANNEX I.1	0.028	103	
13.18 C		0.60	0.83	-	
8 ST PIPE	PASS	ANNEX I.1	0.066	102	
3.63 C		1.25	2.09	-	
9 ST PIPE	PASS	ANNEX I.1	0.118	102	
3.59 C		2.94	3.06	-	
10 ST PIPE	PASS	ANNEX I.1	0.211	102	
3.44 C		6.32	4.39	-	
11 ST PIPE	PASS	ANNEX I.1	0.295	102	
3.44 C		9.44	5.53	-	
12 ST PIPE	PASS	ANNEX I.1	0.248	102	
2.63 C		8.00	4.58	-	